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Research Article

Molecular mapping of α-thrombin (αT)/β2-glycoprotein I (β2GpI) interaction reveals how β2GpI affects αT functions

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β2-Glycoprotein I (β2GpI) is the major autoantigen in the antiphospholipid syndrome, a thrombotic autoimmune disease. Nonetheless, the physiological role of β2GpI is still unclear. In a recent work, we have shown that β2GpI selectively inhibits the procoagulant functions of human α-thrombin (αT; i.e. prolongs fibrin clotting time, t_c, and inhibits αT-induced platelet aggregation) without affecting the unique anticoagulant activity of the protease, i.e. the proteolytic generation of the anticoagulant protein C (PC) from the PC zymogen, which interacts with αT exclusively at the protease catalytic site. Here, we used several different biochemical/biophysical techniques and molecular probes for mapping the binding sites in the αT–β2GpI complex. Our results indicate that αT exploits the highly electropositive exosite-II, which is also responsible for anchoring αT on the platelet GpIbα (platelet receptor glycoprotein Ibα) receptor, for binding to a continuous negative region on β2GpI structure, spanning domain IV and (part of) domain V, whereas the protease active site and exosite-I (i.e. the fibrinogen-binding site) remain accessible for substrate/ligand binding. Furthermore, we provided evidence that the apparent increase in t_c, previously observed with β2GpI, is more likely caused by alteration in the ensuing fibrin structure rather than by the inhibition of fibrinogen hydrolysis. Finally, we produced a theoretical docking model of αT–β2GpI interaction, which was in agreement with the experimental results. Altogether, these findings help to understand how β2GpI affects αT interactions and suggest that β2GpI may function as a scavenger of αT for binding to the GpIbα receptor, thus impairing platelet aggregation while enabling normal cleavage of fibrinogen and PC.

Introduction

β2-Glycoprotein I (β2GpI) is a heavily glycosylated plasma protein (45 kDa), which is synthesized in the liver [1] and represents the major autoantigen in the antiphospholipid syndrome (APS) [2], a severe thrombotic autoimmune disease characterized by arterial and venous thrombosis and recurrent fetal loss [3]. Among the proteins of the coagulation system, the plasma concentration of β2GpI (3–7 μM) is second only to that of fibrinogen (7 μM) and the β2GpI sequence is highly conserved through the animal kingdom, from reptiles to mammals [1]. Intriguingly, whereas the involvement of β2GpI in the pathogenesis of APS is widely accepted, the physiological role of this protein is yet to be firmly established and both pro- and anti-coagulant properties have been reported for this protein [4]. From the structural standpoint, β2GpI is composed of five domains (D-I to D-V), arranged like beads on a string, and displays wide conformational heterogeneity [5]. In particular, two independent X-ray diffraction analyses of β2GpI in the crystal state report an elongated J-shaped structure [6,7], whereas small-angle X-ray scattering in solution indicates that β2GpI predominantly assumes an S-shaped

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conformation [8]. Furthermore, electron microscopy data have recently shown that β2GpI can exhibit either a closed/circular or an open/extended conformation [9]. Interestingly, β2GpI exploits the highly electropositive D-V for binding to negative membrane phospholipids, heparin and cellular receptors [1,4].

Thrombin (αT) is a plasma serine protease with a chymotrypsin-like fold well positioned at the interface between coagulation, inflammation and cell differentiation [10,11]. In hemostasis, αT plays either procoagulant or anticoagulant functions [12], and the equilibrium between these opposite activities is regulated by interactions with other proteins [12,13]. αT procoagulant functions mainly entail conversion of fibrinogen into fibrin and activation of platelets via proteolysis of type 1 protease-activated receptor (PAR1), whereas the anticoagulant functions of αT are essentially related to its ability to proteolytically activate the anticoagulant PC in the presence of thrombomodulin (TM) [10,12]. αT functions are also regulated by sodium ion that binds to a specific site on the protease and enhances its hydrolytic efficiency mainly towards procoagulant substrates [11,14,15]. Thrombin accomplishes most of its activities through the hydrolytic active site and two positively charged exosites (exosite-I and exosite-II) that are located at opposite sides from the catalytic cleft [16]. Exosite-I binds physiological substrates such as fibrinogen and PAR1, cofactors such as TM and non-physiological ligands/inhibitors such as hirudin, hirugen and HD1 aptamer [11,13,17]. Exosite-II is more electropositive than exosite-I and interacts with negatively charged ligands/inhibitors like heparin and heparan sulfate, the prothrombin (ProT, human plasma prothrombin zymogen) F2 fragment, the fibrinogen elongated γ-chain and the platelet receptor glycoprotein Ibα (GpIbα) [11,13,17]. In particular, GpIbα is responsible for the localization of αT on the platelet surface and orients αT for efficient PAR1 cleavage [18].

Very recently, we have shown that β2GpI selectively inhibits the procoagulant functions of αT, by moderately prolonging the fibrin clotting time in plasma coagulation assays and by inhibiting αT-induced platelet aggregation in different experimental settings, i.e. whole blood, gel-filtered platelets and PAR1 cleavage on platelet membranes [19]. Notably, β2GpI did not alter the ability of αT to proteolytically convert the PC zymogen into the active protease (aPC, active protein C), in the presence or in the absence of TM. Hence, we concluded that β2GpI may function as a physiological anticoagulant by inhibiting the procoagulant properties of αT, i.e. fibrin generation and platelet activation, without affecting its unique anticoagulant activity, i.e. aPC generation [19].

In this context, structural information on the β2GpI–αT complex would much help to better understand the role of β2GpI in hemostasis. Unfortunately, despite our efforts during the last few years, we have failed in obtaining crystals of sufficiently good quality for X-ray diffraction studies, probably because of the intrinsic conformational heterogeneity of β2GpI [6–9]. Hence, in the absence of direct structural information, here we combined biochemical/biophysical and molecular modeling techniques for mapping the binding sites in the αT–β2GpI complex and provide a coherent structural model that could help to rationalize at the molecular level the effect of β2GpI on αT functions.

Materials and methods

A detailed description of all the experimental procedures reported in this work is included in the Materials and methods section, which is reported in Supplementary Data.

Reagents

Human plasma αT (EC-3.4.21.5) and ProT were purchased from Haematologic Technologies (Essex Junction, VT, U.S.A.), whereas HD1 aptamer was from Primm (Milan, Italy). Hirugen and *N*α-fluoresceinated hirugen ([F]-hirugen), GpIbα(268–282) peptide, β2GpI(219–232) peptide and its sequence-randomized analog R-β2GpI (219–232), and PAR1(38–60) were synthesized by the solid-phase strategy on a PS3 automated synthesizer (Protein Technologies, AZ, U.S.A.) using fluorenylmethyloxycarbonyl chemistry [20], and thoroughly characterized by RP-HPLC and high-resolution mass spectrometry. Randomization of the β2GpI(219–232) sequence was performed using the Mimotopes software (The Peptide Company, Victoria, Australia). Salts, solvents and reagents were of analytical grade and purchased from Sigma (St. Louis, MO, U.S.A.) or Merck (Germany).

Purification and characterization of β2GpI from human plasma

Plasma samples from blood donors and nonsmoking healthy subjects, with different blood groups (A^+ and O^+), were obtained from the institutional blood bank of the University Hospital of Padua. All subjects gave their informed consent to the present study. Purification of β2GpI from human plasma was carried out following essentially the perchloric acid precipitation method, as recently modified [19,21]. This procedure allowed us

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to obtain highly pure (>98%) preparations of β2GpI, where the major glycosylated component has a molecular mass of 45058.5 ± 1.0 a.m.u. (Supplementary Figure S1).

Spectroscopic methods

Determination of protein concentration

Protein/peptide concentration was determined spectrophotometrically by measuring the absorbance at 280 nm on a Jasco (Tokyo, Japan) V-630 double-beam spectrophotometer, using proper absorptivity values (Supplementary Data). All spectroscopic measurements were carried out at 25°C in 20 mM HEPES (pH 7.4), 0.15 M NaCl (HBS) containing additives as explicitly reported.

Dynamic light scattering

Dynamic light scattering (DLS) measurements were performed on a Zetasizer-Nano-S instrument (Malvern Instruments, U.K.). Each measurement consisted of a single run (15 s). Scattering data were analyzed with the Nano 6.20 software and expressed as percentage intensity size distribution, from which the values of the hydrodynamic diameter (D_H) and the percent polydispersity (%Pd) were extracted. Notably, D_H is the diameter of the hard sphere that diffuses at the same speed as the molecule being measured, whereas %Pd is a parameter describing the width of the particle size distribution of a protein in a given sample (Supplementary Data). The intensity data were then corrected for the relative abundance of each species, estimated from the volume size distribution analysis [22].

Fluorescence

Fluorescence binding measurements were carried out at 25°C in HBS, containing 0.1% PEG-8000, on a Jasco FP-6500 spectrofluorimeter. Aliquots of β2GpI were added to an αT solution and samples were excited at 280 or 295 nm. After subtraction for the corresponding blank spectra of β2GpI, the data points were interpolated with eqn 1 in Supplementary Data to obtain the dissociation constant, K_d, of the complex. In fluorescence competition measurements, aliquots of αT S195A mutant stock solution were incrementally added to a solution of [F]-hirugen. Samples were then excited at 492 nm and the decrease in fluorescence intensity of [F]-hirugen was recorded at 516 nm as a function of αT concentration. The data points were interpolated with eqn 1 in Supplementary Data, as described recently [23].

Surface plasmon resonance

Surface plasmon resonance (SPR) analyses were carried out using a multicycle injection strategy on a dual flow-cell Biacore-X100 instrument (GE Healthcare). Each binding curve was subtracted for the corresponding baseline, accounting for nonspecific binding (<2% of RU_{max}, where RU is response units), and the data were analyzed using the BIAevaluation software. β2GpI was immobilized (1000–5000 RU) on a carboxymethylated-dextran chip (CM5), and increasing concentrations of αT solutions were injected in the mobile phase. In the affinity mode, K_d of the αT–β2GpI complex was obtained by plotting the value of RU_{max}, measured at the steady state, as a function of [αT] and fitting the data points to eqn 2 in Supplementary Data, describing a 1:1 binding model. Measurements were also performed by reversing the interacting system, that is, by immobilizing αT on the sensor chip (500 RU) and injecting β2GpI solutions in the mobile phase [23] (Supplementary Figure S2). To study fibrinogen–β2GpI interaction, fibrinogen-bound CM5 sensor chip was challenged with incremental concentrations of β2GpI and the data points were interpolated with eqn 2 in Supplementary Data. SPR measurements were carried out in HBS, containing 3 mM EDTA and 0.05% polyoxyethylene sorbitan (HBS-EP$^+$).

Preparation of β2GpI and αT derivatives

Fully deglycosylated β2GpI was prepared by treating plasma-purified β2GpI (0.5 mg/ml) with peptide-*N*-glycosidase F (PNGase-F; Roche, Germany) at an enzyme:substrate ratio of 1:50 (w/w) in 0.1 M NH_4HCO_3 (pH 7.9), 0.1% (v/v) Rapigest (Waters, MO, U.S.A.) for 24 h at 37°C. The reaction was quenched with 4% aqueous TFA and analyzed by RP-HPLC and SDS–PAGE (4–12% acrylamide), after Coomassie- and fuxin-based staining and high-resolution mass spectrometry (HR-MS) on a Xevo G2-S Q-TOF instrument (Waters; Supplementary Figure S3A). Nicked β2GpI (β2GpI*) was prepared by treating purified β2GpI (22 μM, 0.5 ml) for 2 h at 37°C with human plasmin at an enzyme:substrate ratio of 1:50 (w/w) in 20 mM Tris–HCl

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(pH 7.5), containing 150 mM NaCl and 0.3 mM $CaCl_2$. β2GpI* was purified by heparin sepharose affinity chromatography and further dialyzed to eliminate excess salt. The purity and chemical identity of β2GpI* was established by SDS–PAGE, RP-HPLC and HR-MS under reducing and non-reducing conditions, yielding mass values fully consistent with the hydrolysis of the single peptide bond K317-T318 in β2GpI domain V (Supplementary Figure S4A).

Recombinant wild-type human αT (rαT), prethrombin-2 (rPre2) and the inactive thrombin S195A mutant (rS195A) were expressed in *Escherichia coli*, as detailed elsewhere [19,24], whereas Ala-mutants at αT exosites were expressed in HEK293 human kidney cells and purified as described recently (Supplementary Figure S5) [25]. β_T-Thrombin (β_TT) was prepared by proteolysis of mature αT (7 μM) with trypsin (35 nM; EC-3.4.21.4, Promega Biosciences, CA, U.S.A.) for 3 h at 37°C [26]. To avoid (auto)proteolytic degradation of β_TT, a 10-fold molar excess of PPACK and tosyl-lysyl chloromethyl ketone was added to selectively inhibit αT and trypsin, respectively. The chemical identity and purity (>85%) of the final β_TT preparation were carefully established by RP-HPLC, SDS–PAGE and high-resolution MS (Supplementary Figure S6).

Kinetics of substrate hydrolysis by αT

Chromogenic substrate S2238

αT-catalyzed hydrolysis of the chromogenic substrate S2238 (D-Phe-Pip-Arg-pNA; Chromogenix, Sweden) was monitored at 25°C by measuring the release of pNA at 405 nm on a Victor-3 microplate reader (PerkinElmer, CA, U.S.A.). The initial velocity of pNA release was plotted against substrate concentration and interpolated with the Michaelis–Menten equation to yield the values of k_{cat} and K_m as fitting parameters [15].

Fibrinogen

The specificity constants, k_{catA}/K_{mA} and k_{catB}/K_{mB}, relative to the release of fibrinopeptides A (FpA) and B (FpB) from fibrinogen by αT were determined by carefully following the method of Shafer and coworkers [27]. Briefly, desalted human fibrinogen, Fb (Sigma; 0.35 μM), in HBS, containing 0.1% PEG-8000, was allowed to react at 37°C with human αT (300 pM). At fixed time intervals, acid-quenched aliquots were centrifuged to eliminate fibrin and uncleaved fibrinogen. The supernatant was lyophilized, dissolved in guanidinium hydrochloride (6 M) and analyzed by RP-HPLC to quantify FpA and FpB release. From the plot of FpA or FpB concentration versus time, the values of the corresponding specificity constants were determined as fitting parameters by interpolating the data points with eqns 6 and 7, respectively, in Supplementary Data.

Type 1 protease-activated receptor(38–60)

The specificity constant, k_{cat}/K_m, relative to the hydrolysis of the synthetic peptide PAR1(38–60), was determined as previously detailed [28]. Briefly, PAR1(38–60) in HBS, containing 0.1% PEG-8000, was reacted at 25°C with human αT (100 pM). At fixed time intervals, acid-quenched aliquots were analyzed by RP-HPLC to quantify the time course release of PAR1(42–60) fragment. From kinetic data, the values of the corresponding specificity constant for αT cleavage were determined as a fitting parameter by interpolating the data points with eqn 8 in Supplementary Data.

Computational methods

Protein structures were visualized with the ViewerPro 4.2 software (Accelerys, Inc., U.S.A.). Electrostatic potential calculations were carried out using the APBS [29] and BLUUES [30] programs. Calculations were performed using a solvent dielectric of 78.14 and a protein dielectric of 2.0 at 298K in 145 mM NaCl. Final electrostatic maps were constructed by subtracting the protein self-energies from the calculated map using the dxmath utility in APBS. Electrostatic calculations and docking simulations did not take into account the contribution of glycan chains. Docking of αT into β2GpI structure was performed using the ClusPro 2.0 software (https://cluspro.bu.edu/login.php) [31], starting from the structure of β2GpI (1c1z) [6] and (D)-phenylalanyl-prolyl-arginyl chloromethyl ketone (PPACK)-inhibited thrombin (1ppb) [16], after removing the inhibitor co-ordinates. The 'van der Waals + electrostatics' option was used for docking simulations.

Turbidimetric measurements

Human fibrinogen (Sigma, cat. F4129) was desalted on an in-house packed Sephadex G10 column (0.8 × 15 cm), eluted (0.3 ml/min) with HBS, containing 0.1% PEG-8000. Fibrin generation was started by adding

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human αT to a solution of freshly desalted fibrinogen. The time course of clot formation was followed by continuously recording at 37°C, on a Jasco V-630 spectrophotometer, the turbidity of the solution, expressed as the decrease in the intensity of transmitted light (or the absorbance increase) at 350 nm, due to scattering of fibrin fibers [32]. The clotting curves were analyzed to extract the values of A_{max} and t_c, where A_{max} is the maximum absorbance value obtained when fibrin generation is complete and t_c is the clotting time determined as the intercept with the time axis of the tangent line to the flex point of the clotting curve.

Results

Probing β2GpI–αT interaction

Dynamic light scattering

A first qualitative indication of αT–β2GpI interaction was obtained by DLS analysis, carried out in triplicate in two different sets of experiments (Figure 1A). The hydrodynamic diameter (D_H) of αT was reproducibly measured as 6.50 ± 1.20 nm, with a %Pd of 18.6, indicating that αT preparation was approximately monodisperse [22]. Purified β2GpI displayed a D_H value of 8.72 ± 2.42 nm with a higher %Pd (25.0), consistent with the extensive glycosylation [1] and high conformational heterogeneity of β2GpI elongated structure [6–9].

When αT and β2GpI were mixed at equimolar concentration (18 μM), a slightly smaller D_H of 7.52 ± 1.56 nm was reproducibly measured, with an even lower %Pd (19.3), compared with that of β2GpI alone (25.0). Notably, a mixture of non-interacting proteins of different size and polydispersity should yield a %Pd value much greater than that of the more polydispersed protein alone [22]. Therefore, the results of DLS analysis are consistent with a picture of αT–β2GpI interaction, whereby αT binds to β2GpI and reduces the size and conformational heterogeneity of the protein by forming a complex in which β2GpI becomes more ordered around αT. This interpretation is consistent with DLS analysis of peptide binding to calmodulin (CaM), where peptide–CaM interaction reduced both the D_H value and %Pd of the protein [33].

Fluorescence spectroscopy

A more quantitative estimate of αT–β2GpI interaction was obtained by measuring the decrease in αT fluorescence intensity at 330 nm after the addition of incremental concentrations of β2GpI (Figure 1B). At saturating β2GpI concentrations (300 nM), a ∼12% decrease in fluorescence was reproducibly measured and interpolation of the data points with eqn 1 in Supplementary Data, describing 1:1 tight binding, yielded a K_d of 50 ± 8 nM. The fluorescence change associated with αT–β2GpI coupling is probably caused by alterations in the

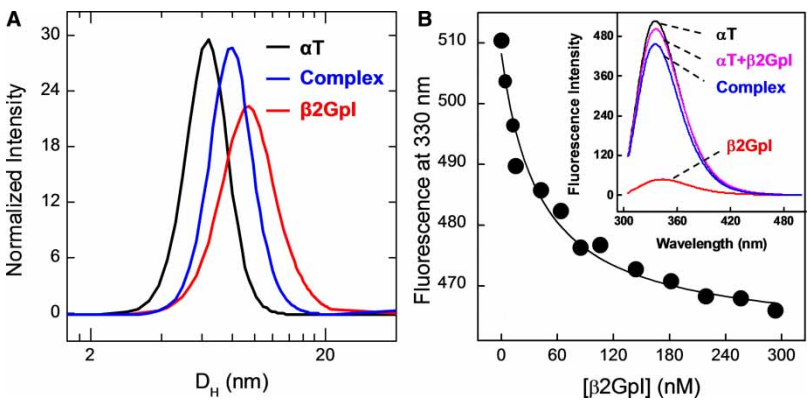


Figure 1. Probing β2GpI–αT interaction by DLS (A) and fluorescence spectroscopy (B).

(**A**) DLS analysis of 4 μM β2GpI (▬) and αT (—) alone or in equimolar mixture (▬). Measurements were conducted by equilibrating samples for 40 min at 25°C in HBS (pH 7.4). (**B**) To a solution of αT (60 nM) in HBS were added aliquots (2–10 μl) of a β2GpI stock solution (11 μM). The samples were excited at 295 nm and emission intensity was recorded at 330 nm. After subtracting the fluorescence of β2GpI alone, the data points were interpolated with eqn 1 in Supplementary Data and a K_d of 50 ± 8 nM was estimated. (Inset) Fluorescence spectra of αT (60 nM, —) and β2GpI (300 nM, ▬) in the isolated and mixed form (▬). The theoretical sum-spectrum is also reported (▬). At each β2GpI concentration, the fluorescence of αT–β2GpI mixture was subtracted for the contribution of β2GpI alone at the same concentration to yield the fluorescence change associated with the formation of the complex.

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environment of Trp-residues, occurring mainly in αT where six of the nine Trp-residues present in the enzyme structure are moderately or highly solvent-exposed and therefore amenable to interact with β2GpI. At variance, Trp-residues in β2GpI are shielded from the solvent (apart from Trp316 in domain V) and their fluorescence is strongly quenched (Figure 1B, inset) by stacking interactions onto internal disulfide bridges [6,7,34].

Surface plasmon resonance

SPR measurements were performed by immobilizing either β2GpI or αT on the sensor chip (Figure 2), as detailed in Supplementary Data. In the first set of experiments, β2GpI was covalently immobilized on a CM5 chip, via the amine coupling chemistry, and then incremental concentrations of αT were injected in the mobile phase (Figure 2A,B). As described in Supplementary Figure S7, it is expected that most of β2GpI molecules are anchored onto the sensor chip through the lower face of domain V, containing >3-fold higher percentage of reactive Lys-residues, compared with the rest of the protein [1]. SPR data were analyzed according to the

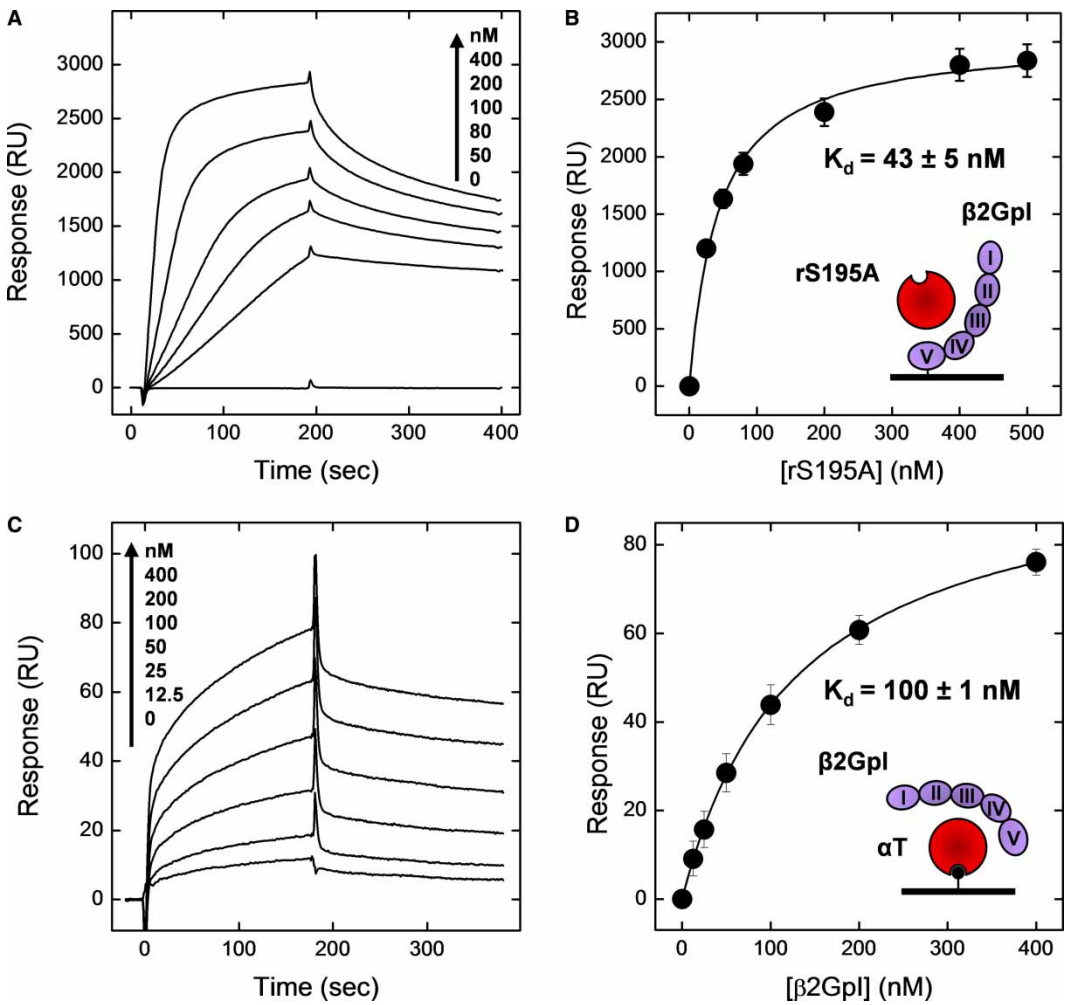


Figure 2. Probing β2GpI–αT interaction by SPR.

(**A** and **B**) Binding of rS195A to β2GpI immobilized on a CM5 sensor chip. (**A**) Sensograms relative to rS195A binding. (**B**) Plot of RU_{max} versus rS195A concentration (●). Fitting of data points with eqn 2 in Supplementary Data yielded a K_d value of 43 ± 5 nM. (**C** and **D**) Binding of β2GpI to αT, immobilized on a NeutrAvidin-coated CM1 sensor chip (see Materials and methods). (**C**) Sensograms relative to β2GpI binding. (**D**) Plot of RU_{max} versus β2GpI concentration (●). Fitting of data points with eqn 2 in Supplementary Data yielded a K_d value of 100 ± 1 nM. All measurements were carried out at 25°C in HBS-EP$^+$ (pH 7.5). Each data point represents the average of three different runs performed on different plasma samples. The standard deviations are shown as error bars.

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one-site binding model (see eqn 2 in Supplementary Data), yielding a K_d of 43 ± 5 nM, identical with that estimated by fluorescence measurements in solution. In the second set of experiments, the interacting system was reversed, i.e. αT was covalently reacted at the active site with the bivalent reagent biotinyl-PPACK and anchored to a NeutrAvidin-coated C1 chip through the biotin–avidin capture strategy, as recently described (Figure 2C,D) [23]. Incremental concentrations of β2GpI were then injected in the mobile phase and, after data fitting as above, only a slightly higher K_d of 100 ± 1 nM was determined. The results of DLS, fluorescence and SPR analyses concurrently indicate that β2GpI in solution exists in a conformation fully competent for binding to αT.

The role of ionic interactions and the effect of salt type in β2GpI–αT binding

The role of ionic interaction in αT–β2GpI complex formation was probed by measuring the affinity of αT for β2GpI at increasing NaCl concentrations. The Debye–Hückel theory, indeed, predicts that if the binding of two interacting proteins is driven by ionic interactions, then increasing salt concentrations should weaken the affinity through unspecific screening of charges on the interacting protein surfaces [35]. SPR data in Figure 3A indicate that this is actually the case, as the affinity of the inactive S195A mutant thrombin for immobilized β2GpI decreased by 32-fold, going from 50 to 400 mM NaCl. Notably, the S195A mutant can be considered as a pseudo-wild-type enzyme. Indeed, this mutant is inactive due to the replacement of the catalytic Ser195 with Ala, but it retains the molecular recognition properties of natural αT [19,24]. A more quantitative estimate of the contribution of ionic interactions was obtained as reported by Bode and coworkers for studying the salt-dependent interaction of αT with the bivalent inhibitor hemadin [36]. The standard free energy change of binding, $\Delta G^\circ_b = RT \cdot \ln K_d$, was partitioned into an ionic (ΔG°_I) and a nonionic component (ΔG°_ϕ), as follows: $\Delta G^\circ_b = \Delta G^\circ_I + \Delta G^\circ_\phi$ [36]. The values of ΔG°_b, obtained from experimental dissociation constant values (K_d), were plotted against the square root of the ionic strength (I) (Figure 3B) and the data points were interpolated with eqn 3 in Supplementary Data, yielding $\Delta G^\circ_\phi = -8.2 \pm 0.1$ kcal/mol and $\Delta G^\circ_{I^\circ} = -7.3 \pm 0.9$ kcal/mol. In this analysis, ΔG°_ϕ is the value of ΔG°_b extrapolated at $I \to \infty$ and $\Delta G^\circ_{I^\circ}$ is the value of ΔG°_I extrapolated at $I \to 0$. These data suggest that ionic interactions make up to 47% of the total binding energy at zero ionic strength ($\Delta G^\circ_b = -15.5$ kcal/mol), thus confirming the importance of charged interactions in αT–β2GpI binding. The effect of salt was further quantified according to the counter-ion condensation (CC) theory of protein–polyelectrolyte interaction, following the method that Olson et al. [37] used for studying αT binding to heparin (Figure 3C). According to the CC theory, the binding of a positively charged protein to a negative polyelectrolyte, in the presence of NaCl, can be regarded as an ion exchange-like process, whereby protein binding is driven by the entropically favorable release of cations (i.e. Na^+) bound to the polyelectrolyte, whereas the displacement of anions (i.e. Cl^-) from the positive protein only marginally contributes. The values of $\log K_d$ are plotted versus $\log[Na^+]$ and the data points were interpolated with the linear equation $\log K_d = A_0 + \Gamma_{salt} \cdot \log[Na^+]$, where A_0 is the value of $\log K_d$ extrapolated at $\log[NaCl] = 0$ and Γ_{salt} is the slope of the straight line. Notably, A_0 represents the nonionic contribution to binding, as it is calculated at 1 M NaCl, where the contribution of ionic interactions to the binding energy is virtually zero due to charge-screening effects. On the other hand, Γ_{salt} represents a thermodynamic measure of the effect of salt concentration on binding equilibria. Assuming that this effect is solely due to ionic interactions, then Γ_{salt} gives the number of Na^+ ions released upon protein–protein interaction and can be expressed as $\Gamma_{salt} = Z \cdot \psi$, where Z is the number of ionic interactions involved in binding and ψ is the fraction of a cation bound to the polyelectrolyte per unit charge and depends on the charge density of the polyelectrolyte (ψ = 0.8 for heparin). From the parameters of the straight line in Figure 3C, it was possible to estimate the nonionic contribution to the binding energy of αT–β2GpI interaction as $\Delta G^\circ_\phi = -7.5$ kcal/mol, comparable with that determined by the Debye–Hückel theory, and infer that two intermolecular salt bridges can be formed in the αT–β2GpI complex. The latter value is close to that determined by Beglova and coworkers for the binding of the recombinant β2GpI domain V to the heparin-derived pentasaccharide fondaparinux ($Z = 2.5$) [38].

αT is a sodium-activated enzyme, whereby Na^+ binds to a specific site (K_d = 18 mM at 25°C) and induces structural changes in αT [15,39], which more efficiently cleaves procoagulant substrates, such as fibrinogen and PAR1, than the anticoagulant substrate PC zymogen [12]. The possible role of salt type was investigated by measuring the affinity of rS195A for immobilized β2GpI at constant ionic strength (0.2 M) in the presence of salts (i.e. ChCl, NaCl, LiCl and NaF) having different cation ($Ch^+ >> Na^+ > Li^+$) and anion ($Cl^- > F^-$) size. SPR data in Figure 3D indicate that with ChCl and LiCl, the affinity remained essentially constant (within the error

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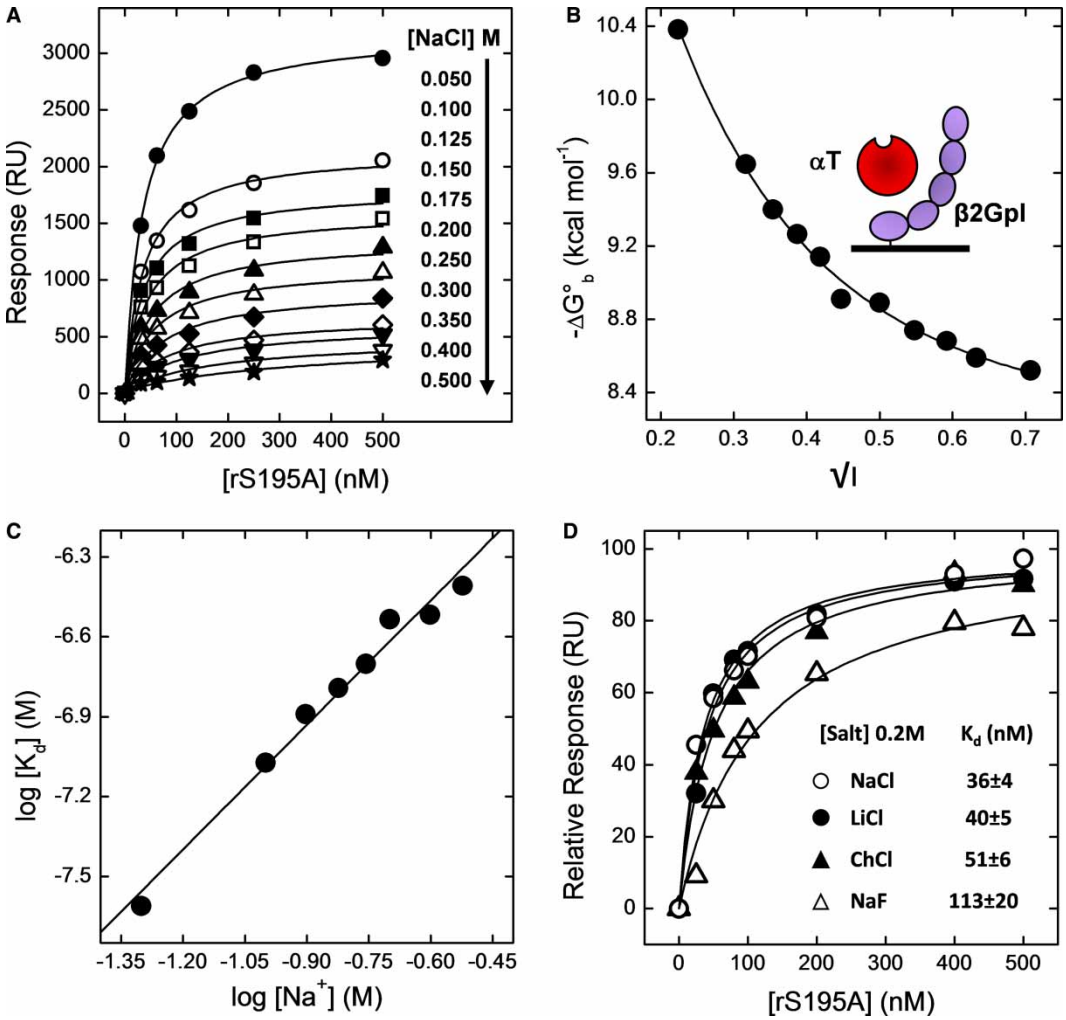


Figure 3. Effect of ionic strength and salt type on β2GpI–αT interaction.

(**A**) SPR analysis of the effect of increasing NaCl concentration on the binding of rS195A to immobilized β2GpI. The binding data were interpolated with eqn 2 in Supplementary Data to estimate the K_d value at each [NaCl]. (**B**) Plot of the binding energies (ΔG_b°) of rS195A–β2GpI interaction against the square root of the ionic strength, $\sqrt{I}$. Here, ΔG_b° values were calculated from the K_d values determined in (**A**): $\Delta G_b^\circ = RT \cdot \ln K_d$. The data points (●) were interpolated with eqn 3 in Supplementary Data, with best-fit parameters: $\Delta G_\phi^\circ = -8.2 \pm 0.1$ kcal/mol, $\Delta G_{I^\circ}^\circ = -7.3 \pm 0.9$ kcal/mol and $\alpha = 3.1 \pm 0.4$. (**C**) Salt dependence of the affinity of αT for β2GpI, as determined by SPR measurements. The data points were fitted to the equation $\log K_d = A_0 + \Gamma_{salt} \cdot \log[Na^+]$, with best-fit parameters $A_0 = -5.53$M and $\Gamma_{salt} = 1.6 \pm 0.2$. (**D**) Effect of the salt type on the affinity of rS195A for immobilized β2GpI. Measurements were carried out at constant concentration (i.e. 0.2 M) of different salts. After data fitting with eqn 2 in Supplementary Data, the reported K_d values were obtained. All measurements were carried out at 25° C in HBS-EP$^+$ (pH 7.5).

of the SPR technique), compared with NaCl, and was only slightly reduced with NaF, thus arguing against a specific contribution of a cation/anion effect in αT–β2GpI interaction.

The role of glycosylation in β2GpI–αT interaction

Both αT and β2GpI are *N*-glycosylated proteins, with the carbohydrate chain moiety contributing to ∼7 and 20% of the total protein weight, respectively [1,16]. To study the role of glycosylation in αT–β2GpI interaction, the corresponding deglycosylated proteins were produced and their affinities were measured by SPR (Figure 4). Deglycosylated β2GpI (d-β2GpI) was obtained in high yields (>85%) by treating plasma-purified β2GpI with

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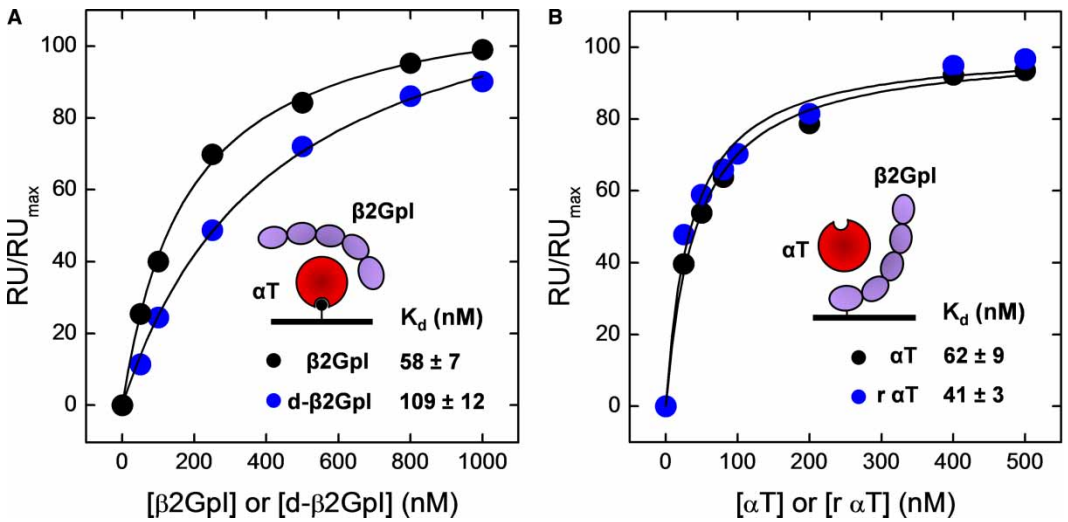


Figure 4. Effect of glycosylation on β2GpI–αT interaction.

(**A**) Effect of deglycosylation of β2GpI. SPR analysis of the binding of β2GpI (●) and d-β2GpI (●) to immobilized natural/glycosylated αT. (**B**) Effect of deglycosylation of αT. SPR analysis of the binding of αT (●) and r-αT (●) to immobilized natural/glycosylated β2GpI. All measurements were carried out at 25°C in HBS-EP$^+$ (pH 8.0). Here, K_d values were obtained by fitting the data points with eqn 2 in Supplementary Data.

PNGase-F glycosidase (Supplementary Figure S3A). For αT, the deglycosylated form (d-αT) was obtained as the expression product of human αT cDNA in *E. coli*, after disulfide oxidative refolding and ecarin activation (Supplementary Figure S3B) [19,24]. SPR data indicate that deglycosylation of β2GpI only marginally (i.e. <2-fold) reduced the affinity of the protein for immobilized natural αT (Figure 4A). Likewise, deglycosylation

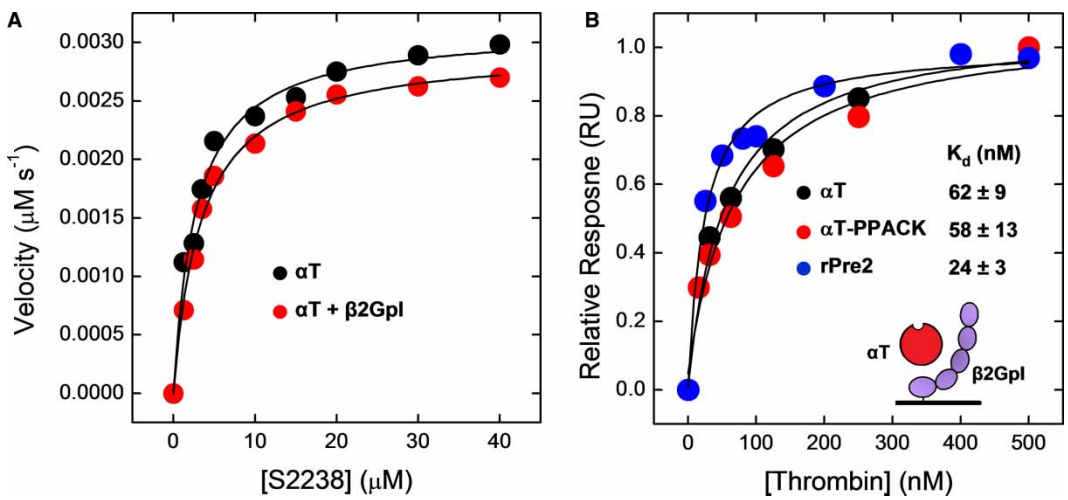


Figure 5. The role of αT active site in β2GpI–αT interaction.

(**A**) Effect of β2GpI on the hydrolytic activity of αT. The initial rate of pNA release from S2238 by αT (50 pM) was measured at 405 nm in HBS, 0.1% PEG at 25°C in the absence (●) or in the presence (●) of 4 μM β2GpI. The data points were interpolated with the Michaelis–Menten equation, with best-fit parameters: −β2GpI, $k_{cat} = 62 \pm 5$ s^{-1}, $K_m = 2.8 \pm 0.3$ μM, $k_{cat}/K_m =$ 22.4 μM^{-1} s^{-1}; +β2GpI, $k_{cat} = 60 \pm 4$ s^{-1}, $K_m = 3.4 \pm 0.3$ μM, $k_{cat}/K_m = 21.5$ μM^{-1} s^{-1}. (**B**) Effect of active-site occupation/perturbation on αT thrombin binding to β2GpI. SPR analyses of the binding of natural αT (●), αT-PPACK (●) and recombinant Pre2 zymogen (●). All measurements were carried out at 25°C in HBS-EP$^+$ (pH 8.0). The data points were interpolated with eqn 2 in Supplementary Data, yielding the corresponding K_d values, as reported.

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of αT did not alter the affinity for the immobilized natural β2GpI (Figure 4B). These findings provide evidence that (de)glycosylation does not significantly influence β2GpI–αT interaction.

Mapping of the binding sites on αT

The role of the αT active site and recognition exosites in β2GpI binding was assessed by measuring the effect of β2GpI on the efficiency of substrate hydrolysis by αT or by selectively introducing incremental structural perturbations at a given site of the protease and then measuring the affinity change in the perturbed αT species for β2GpI. A strong decrease in affinity was taken as an indication that the perturbed site was relevant for the interaction of αT with β2GpI.

Active site

The involvement of the αT active site was directly probed by measuring the efficiency of hydrolysis of the specific chromogenic substrate S2238 in the presence or absence of β2GpI (4 μM). The data shown in Figure 5A indicate that β2GpI did not alter the kinetic constants, K_m and k_{cat}, for substrate hydrolysis. These results provide straight evidence that β2GpI did not affect the accessibility nor the efficiency of αT catalytic machinery and strongly suggest that the protease active site is not involved in binding to β2GpI. The role of the αT active site in β2GpI interaction was further probed by SPR, challenging a β2GpI-loaded sensor chip with PPACK-inhibited αT (αT-PPACK) or with the recombinant prethrombin-2 (Pre2) zymogen. Notably, PPACK covalently reacts with Ser195 and His57 in the catalytic pocket and extensively penetrates in the S1, S2 and S3 substrate-binding sites [16]. On the other hand, Pre2 structure shows major perturbations in the Na^+-binding site and activation domain, leading to disruption of the catalytic site that becomes also occluded by the partial collapse of the surrounding insertion loops, whereas the exosites remain functional [40]. SPR data (Figure 5B) show that both αT-PPACK and Pre2 retain the affinity of native αT for immobilized β2GpI, thus indicating that neither the occupancy of αT catalytic cleft nor disruption of the protease active site compromised binding to β2GpI.

Exosites

Binding of thrombin Ala-mutants to β2GpI

Twelve single Ala-mutants of αT were produced in HEK293 human kidney cells and purified to homogeneity (Supplementary Figure S5A) [25], while their affinity for immobilized β2GpI was determined by SPR. Our data indicate that Ala-shaving of hydrophobic (Phe/Leu) or charged (Arg/Lys) amino acids at exosite-I only slightly altered, either positively or negatively, the K_d for the binding to β2GpI (<1.5-fold; Supplementary Figure S5B). Similar results were obtained with Arg/Lys → Ala-mutants at exosite-II, and the only substitution that caused a significant change in the affinity was D178A, which rather surprisingly enhanced by ∼4-fold the affinity of αT for β2GpI compared with the natural enzyme (Supplementary Figure S5B). For a correct interpretation of these data, it should be considered that the net effect of a charged-to-neutral amino acid exchange on ligand–protein interaction is the result of compensatory (often opposing) effects that the mutation has on the ligand-binding energetics, but also on the local protein environment at the mutation site [41]. These effects are difficult to predict and can be rationalized only when a reliable model of the ligand–protein complex is available. The poor sensitivity of αT–β2GpI binding to Ala-mutations, however, does not rule out the possibility that αT exosites are involved in β2GpI binding. Instead, these data suggest that the perturbation introduced by single-charge ablation might be too small to cause large affinity loss, as already observed in other thrombin interacting systems, where multiple Arg/Lys → Ala or even charge-reversal mutations (Arg/Lys → Asp/Glu) are needed to elicit significant changes in affinity [36,42,43]. Electrostatic potential calculations, indeed, show that this is actually the case, as the estimated change of surface potential after mutation is small, on average 6.9 ± 2.8% for exosite-I and 6.3 ± 2.6% for exosite-II (Supplementary Figure S5C).

Effect of exosite-specific ligands on αT–β2GpI interaction

Given the relative insensitivity of exosite electrostatics to point mutations, we decided to more strongly perturb αT through the binding of negatively charged ligands that electrostatically couple with and cover a much larger area on the enzyme surface. Hence, we devised an SPR competition experiment (Figure 6) where S195A (250 nM) was incubated with increasing concentrations of a given exosite-specific ligand and then injected over the β2GpI-sensor chip. The C-terminal peptide segment 268–282 of the αT platelet receptor, GpIbα, was used

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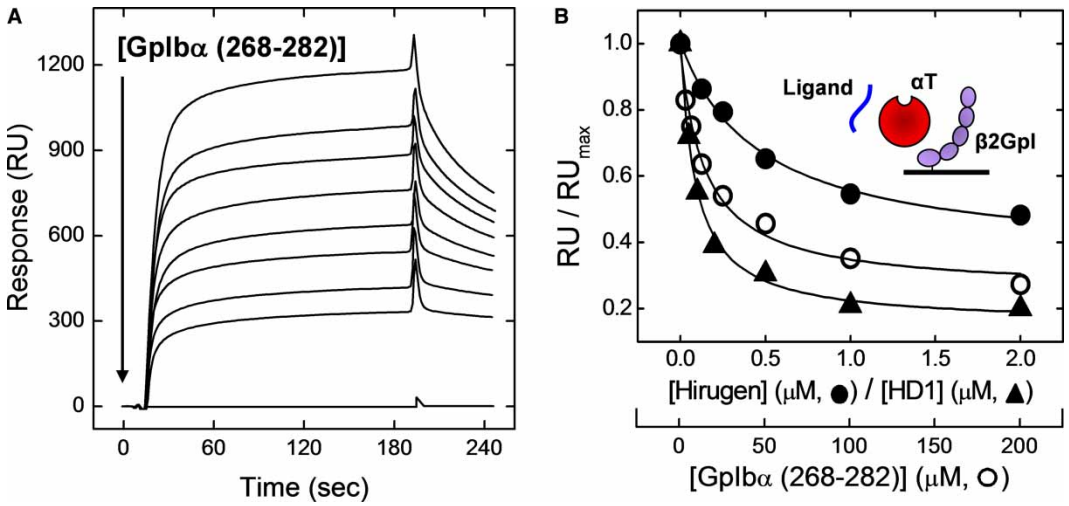


Figure 6. Effect of exosite occupation on the affinity of αT for β2GpI.

(**A**) Sensograms relative to the binding of rS195A (250 nM) to immobilized β2GpI in the presence of increasing concentrations of the exosite-II-specific ligand GpIbα(268–282). (**B**) Plot of RU/RU$_{max}$ values at increasing concentrations of Hirugen (●), GpIbα(268–282) (○) and HD1 aptamer (▲). The data points were interpolated with eqn 4 in Supplementary Data, describing the competition of two ligands for the same receptor site. The following best-fit parameters were obtained: Hirugen, $K_d^{\alpha T} = 15 \pm 3$ nM and $K_d^{L} = 1.5 \pm 0.4$ μM; HD1, $K_d^{\alpha T} = 11 \pm 2$ nM and $K_d^{L} = 1.7 \pm 0.1$ μM; GpIbα(268–282), $K_d^{\alpha T} = 12 \pm 3$ nM and $K_d^{L} = 0.5 \pm 0.1$ μM, where $K_d^{\alpha T}$ is the K_d for the binding of αT to the immobilized β2GpI, whereas K_d^{L} is the K_d for the binding of αT to the ligand peptide (L) in the mobile phase. All measurements were carried out at 25°C in HBS-EP$^+$ (pH 7.4).

as a specific exosite-II ligand ($K_d = 3$ μM) [43,44], whereas hirugen ($K_d = 1$ μM) and HD1 aptamer ($K_d = 0.5$ μM) were used as exosite-I binders [17]. Notably GpIbα(268–282) reduced up to 70% the RU$_{max}$ of αT binding to β2GpI, in a dose dependent manner (Figure 6A,B). This effect was taken as an indication that the

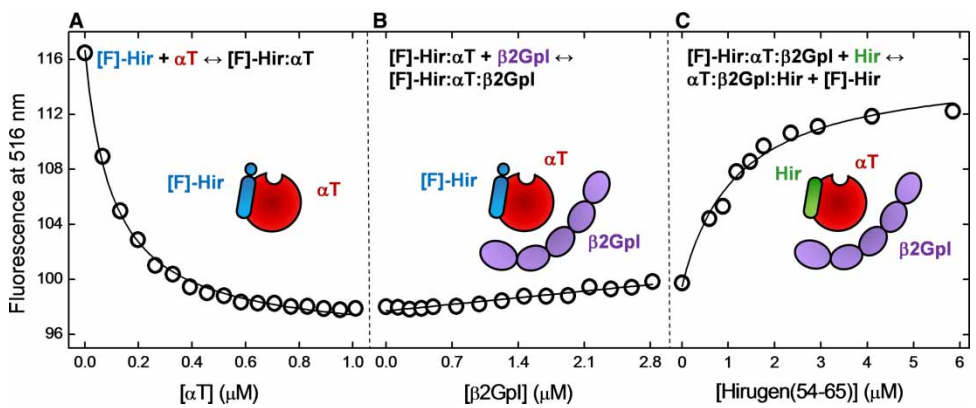


Figure 7. Competition of fluoresceinated hirugen and β2GpI for αT exosite-I.

Fluorescence competition measurements for the binding of αT and β2GpI to [F]-hirugen. (**A**) Binding of rS195A to [F]-hirugen (60 nM). The data points were interpolated with eqn 1 in Supplementary Data, yielding $K_d = 30 \pm 8$ nM. (**B**) Displacement of [F]-hirugen from the complex with rS195A, as obtained in **A**, by adding incremental concentrations of β2GpI. A moderate increase in fluorescence (~10%) was observed, suggestive of only a partial displacement of [F]-hirugen by β2GpI. (**C**) Displacement of [F]-hirugen from rS195A by adding incremental concentrations of *unlabeled* hirugen. A marked increase in fluorescence (~90%) was measured as the result of complete displacement from αT exosite-I of [F]-hirugen that was still bound to the enzyme in the presence of β2GpI. Samples in HBS, containing 0.1% PEG-8000, were excited at 25°C at 492 nm and the fluorescence of [F]-hirugen, bound to or released from S195A, was recorded at 516 nm (Supplementary Data).

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peptide in solution can compete with immobilized β2GpI for interacting with αT and that the protease exosite-II is involved in binding to β2GpI. Intriguingly, the addition of incremental concentrations of exosite-I-specific ligands, i.e. hirugen and HD1, also significantly reduced the RU_{max} of αT binding to immobilized β2GpI (Figure 6B). The data were analyzed using the SPR competition model recently reported by Gamsjaeger et al. [45], describing the competition of two ligands (one of which is immobilized on the sensor chip) for the same receptor in the mobile phase. Interpolation of the data points with eqns 4 and 5 in Supplementary Data yielded values of $K_d\alpha^T$ (i.e. the K_d for the binding of αT to chip-bound β2GpI) and K_d^L (i.e. the K_d for the binding of αT to an exosite-specific ligand in the mobile phase) that were comparable with those estimated by direct binding measurements, thus supporting the validity of the competition model used.

At a first glance, these findings would suggest that, besides exosite-II, also exosite-I participates in β2GpI interaction. Nevertheless, our data can be also explained by the negative allosteric linkage existing between αT exosites, whereby the binding of a ligand at exosite-I can perturb long-range exosite-II structure and decrease the affinity of that site for a specific ligand and vice versa [46–48]. Hence, the decrease of SPR signal measured with hirugen or HD1 does not imply a direct involvement of exosite-I in β2GpI interaction and might be caused by an indirect effect that the binding of these ligands at exosite-I have on the affinity of exosite-II for β2GpI. Clearly, this further complicates a straightforward interpretation of the data reported in Figure 6B and prompted us to devise alternative strategies for more precisely addressing the role of αT exosites.

Displacement of fluorescein-labeled hirugen ([F]-hirugen) from αT exosite-I

To reduce the ambiguity in interpreting the effect of exosite-I blockage in β2GpI binding, the ability of β2GpI to displace [F]-hirugen from αT exosite-I was investigated (Figure 7). In the first step, [F]-hirugen was titrated with S195A, and a K_d of 30 ± 8 nM was estimated, consistent with previous work [17,23,46]. In the second step, increasing concentrations of β2GpI were added to the solution of S195A, previously saturated with [F]-hirugen. Only 12% of the initial fluorescence of [F]-hirugen was recovered at saturating β2GpI concentration (3 μM). In the third step, [F]-hirugen still bound to S195A was displaced by adding increasing concentrations of unlabeled hirugen. As expected, the initial fluorescence characteristic of free [F]-hirugen was almost fully recovered. Our results indicate that naïve hirugen, *but not* β2GpI, is capable of displacing the majority of [F]-hirugen molecules from thrombin exosite-I, thus providing strong, albeit indirect, evidence that αT exosite-I is not involved in binding to β2GpI.

Binding of thrombin derivatives to β2GpI

The role of exosites in αT–β2GpI interaction was further investigated by SPR using αT derivatives, i.e. ProT and β_TT, having the exosites variably compromised (Figure 8A). Notably, both ProT and β_TT have been already exploited for mapping the αT-binding site in other interacting systems [13,17,23]. ProT is the physiological inactive precursor of mature αT and is composed of a γ-carboxyglutamic acid (Gla) domain, two kringle domains and a protease domain. As reported for Pre2, the Na^+-binding site, the activation domain and the insertion loops surrounding the catalytic pocket in ProT are compromised, while exosite-I is only slightly altered compared with αT structure [49]. At variance with Pre2, exosite-II reactivity in ProT is abolished due to masking of the kringle-2 domain. β_TT is a non-physiological nicked species of αT (Supplementary Figure S6), resulting from limited proteolysis of the mature enzyme with trypsin at the single peptide bond Arg77a–Asn78 in the exosite-I, which is selectively disrupted in β_TT, whereas the catalytic site and exosite-II remain fully functional [26]. SPR data (Figure 8B) indicate that masking of exosite-II reactivity, as in ProT, causes a drop (7-fold) in the affinity of αT for immobilized β2GpI, whereas killing of exosite-I, as in β_TT, did not alter the binding strength. These results further support our conclusion that exosite-II is the hot spot on αT for binding to β2GpI.

Effect of β2GpI on fibrinogen and PAR1 hydrolysis

αT exploits exosite-I for binding to either fibrinogen or PAR1 and for properly orienting these substrates in the protease active site for efficient cleavage [11,13,17]. Hence, to probe the effect of β2GpI on the functional state of exosite-I, beyond simple binding, we evaluated whether physiological β2GpI concentrations (4 μM) alter the efficiency with which αT releases fibrinopeptides (FpA and FpB) from fibrinogen (Figure 9A) and cleaves PAR1(38–60) (Figure 9B). Notably, the latter peptide reproduces the substrate-binding properties of the extracellular domain of PAR1 on platelets, as it contains both the exosite-I binding sequence and the scissile bond R^{41}-S^{42} for αT [28]. The kinetic data relative to fibrinopeptide generation were analyzed within the framework

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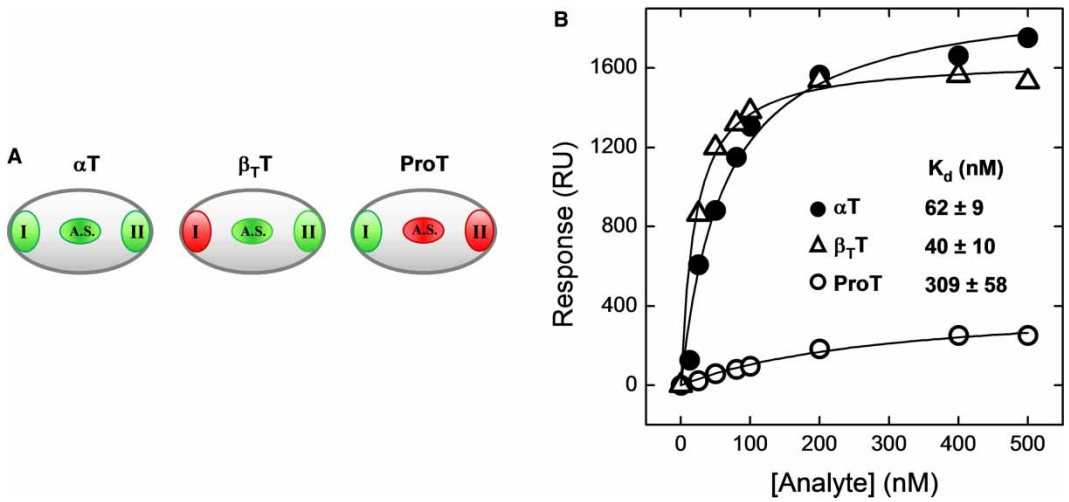


Figure 8. Effect of selective perturbation of exosites on the affinity of αT for β2GpI.

(**A**) Schematic representation of the protease domain of mature αT, β_TT and ProT. The active site (A.S.) and exosite-I (I) and exosite-II (II) are colored according to the conformational/functional state they assume in the different thrombin derivatives, compared with αT (see text); green: unperturbed or only slightly perturbed; red: heavily perturbed. (**B**) SPR analysis of the binding of αT, β_TT and ProT to a β2GpI-sensor chip. The values of RU_{max} were plotted versus the concentration of thrombin derivatives, and the data points were interpolated with eqn 2 in Supplementary Data, yielding the corresponding K_d values, as indicated. All measurements were carried out at 25°C in HBS-EP$^+$ (pH 7.4).

of the Shafer's model [27], whereby a highly specific cleavage at the Aα chain leads to the rapid release of FpA and formation of fibrin I monomers that aggregate to form fibrin I protofibrils. A second cleavage by αT at the Bβ chain of the fibrin I monomers then leads to the slower release of FpB and formation of fibrin II protofibrils, which then aggregate to form the fibrin clot. The values of the specificity constants (k_{cat}/K_m) for the release of FpA and FpB, extracted from the kinetic data in Figure 9A, were found to be identical with those reported earlier [27] and, more importantly, independent of β2GpI addition. Likewise, the k_{cat}/K_m value of PAR1(38–60) hydrolysis by αT was identical with that previously estimated [18,28] and remained essentially constant after the addition of 4 μM β2GpI (Figure 9B). The relative insensitivity to β2GpI of αT-mediated cleavage of two exosite-I binders like fibrinogen and PAR1 provides evidence that exosite-I plays a minor (if any) role in αT–β2GpI recognition.

Mapping of the binding sites on β2GpI

Considering that formation of the αT–β2GpI complex is significantly contributed by ionic interactions (Figure 3) and that the highly positive αT exosite-II is involved in binding, we sought for large negative spots on β2GpI structure as putative binding sites for αT. Domains I, III and V were all discarded as they are positively charged at the plasma pH (pI: 8.8, 8.2 and 8.9). Although highly basic, D-V displays an asymmetric distribution of charges, whereby the upper surface is predominantly negative, whereas the lower surface is strongly positive (see Figure 11A). The possible involvement of D-II (pI: 4.7) was also ruled out as the six acidic residues present in the sequence are dispersed on the domain surface, do not form a sufficiently large negative patch and are partially compensated by basic amino acids nearby [6,7]. At variance with D-II, D-IV (pI: 5.6) contains the segment ^{222}**D**GP**EE**I**E**^{228}, which forms a large continuous region of uncompensated negative charges on the β2GpI surface [6,7], and thus, it may function as a linear epitope for αT binding.

The peptide β2GpI(219–232), ^{219}YSL**D**GP**EE**I**E**STKL^{232}, was synthesized and its affinity for αT was measured by fluorescence spectroscopy. As already observed with full-length β2GpI (Figure 1B), the synthetic peptide β2GpI(219–232) reduced αT fluorescence intensity (Figure 10A), even though to a lower extent compared with β2GpI (Figure 10B), and bound to the protease with similar affinity ($K_d = 85 \pm 15$ nM). However, the binding was not specific, as the synthetic peptide analog R-β2GpI(219–232), having the randomized sequence **D**LYIPSL**E**GKT**E**S**E**, reproduced the fluorescence changes induced by the natural peptide

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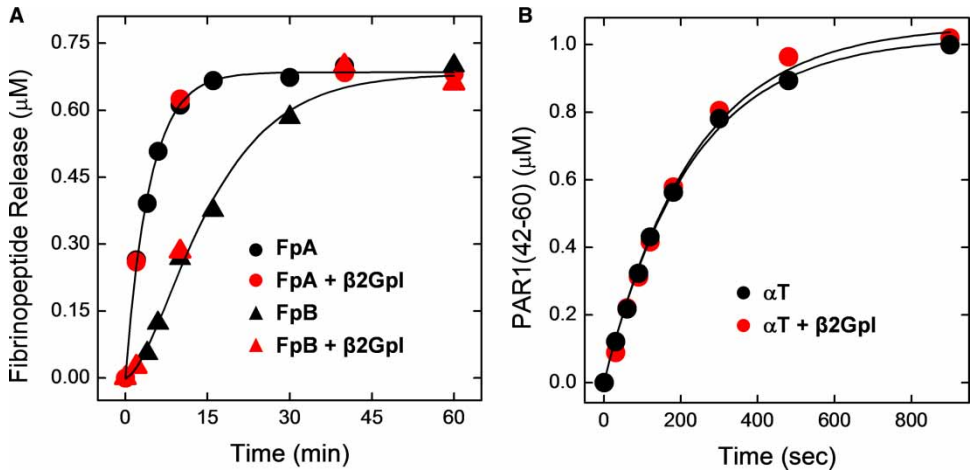


Figure 9. Effect of β2GpI on the efficiency of fibrinogen and PAR1 hydrolysis by αT.

(**A**) Time course analysis of fibrinopeptide release. Release of FpA (●, ●) and FpB (▲, ▲) from human fibrinogen (0.35 μM) by αT (0.3 nM) in the absence (●, ▲) or in the presence (●, ▲) of 4 μM β2GpI. Experiments were carried out at 37°C in HBS. Fibrinopeptides were quantified by RP-HPLC. Interpolation of the data points relative to FpA and FpB release with eqn 6 or 7 in Supplementary Data, respectively, yielded kinetic constant values as $k' = 3.74 \pm 0.12 \times 10^{-3}$ s^{-1} and $k'' = 1.11 \pm 0.38 \times 10^{-3}$ s^{-1}, from which the values of the apparent specificity constants were obtained as $k_{catA}/K_{mA} = 12.5 \pm 0.6$ s^{-1} μM^{-1} and $k_{catB}/K_{mB} = 3.7 \pm 0.8$ s^{-1} μM^{-1}. (**B**) Time course analysis of PAR1(38–60) cleavage by αT in the absence (●) or in the presence (●) of 4 μM β2GpI. The time course release of the PAR(42–60) fragment was quantified by RP-HPLC and the data points were fitted with eqn 8 in Supplementary Data to yield the following k_{cat}/K_m values: −β2GpI, $k_{cat}/K_m = 4.5 \pm 0.2 \times 10^7$ M^{-1} s^{-1}; +β2GpI, $k_{cat}/K_m = 4.3 \pm 0.3 \times 10^7$ M^{-1} s^{-1}.

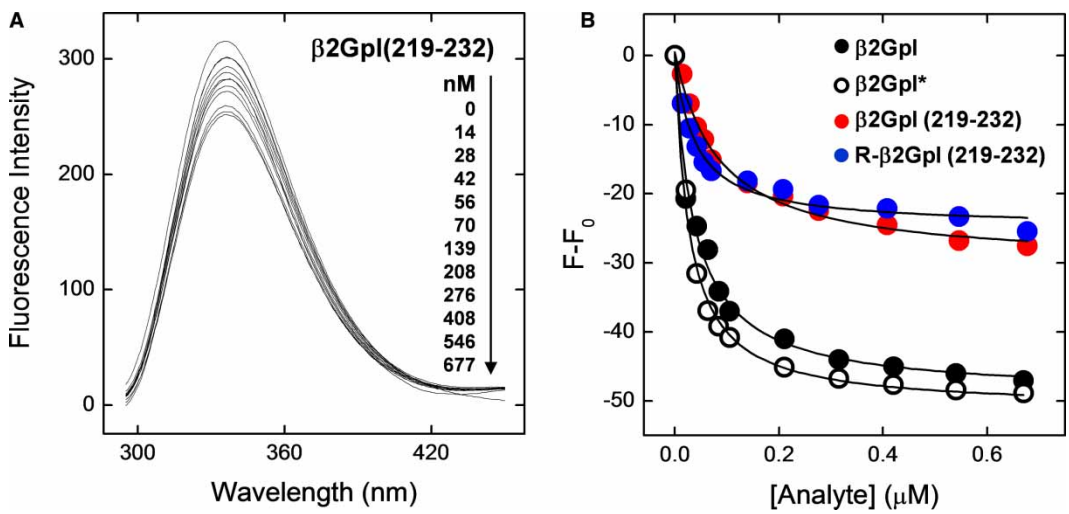


Figure 10. Binding of β2GpI(219–232) and nicked β2GpI* to αT.

(**A**) Fluorescence spectra of αT in the presence of increasing concentrations of the β2GpI(219–232) peptide. At each β2GpI (291–232) concentration, the fluorescence of the αT–β2GpI peptide mixture was subtracted for the contribution of the β2GpI peptide alone at the same concentration, to yield the fluorescence change associated with complex formation. (**B**) Plot of the fluorescence change of αT versus the concentration of the peptide β2GpI(219–232) (●), the sequence-randomized peptide R-β2GpI(219–232) (●), full-length β2GpI (●) and the plasmin-nicked species β2GpI* (○). To a solution of αT (60 nM) were added aliquots (2–5 μl) of analyte, i.e. β2GpI or β2GpI peptides. The samples were excited at 295 nm and the emission intensity was recorded at 330 nm. After subtracting the fluorescence of the analyte alone, the data points were interpolated with eqn 1 in Supplementary Data to yield K_d values: β2GpI(219–232), $K_d = 85 \pm 15$ nM; R-β2GpI(219–232), $K_d = 49 \pm 6$ nM; β2GpI, $K_d = 50 \pm 8$ nM; β2GpI*, $K_d = 45 \pm 5$ nM.

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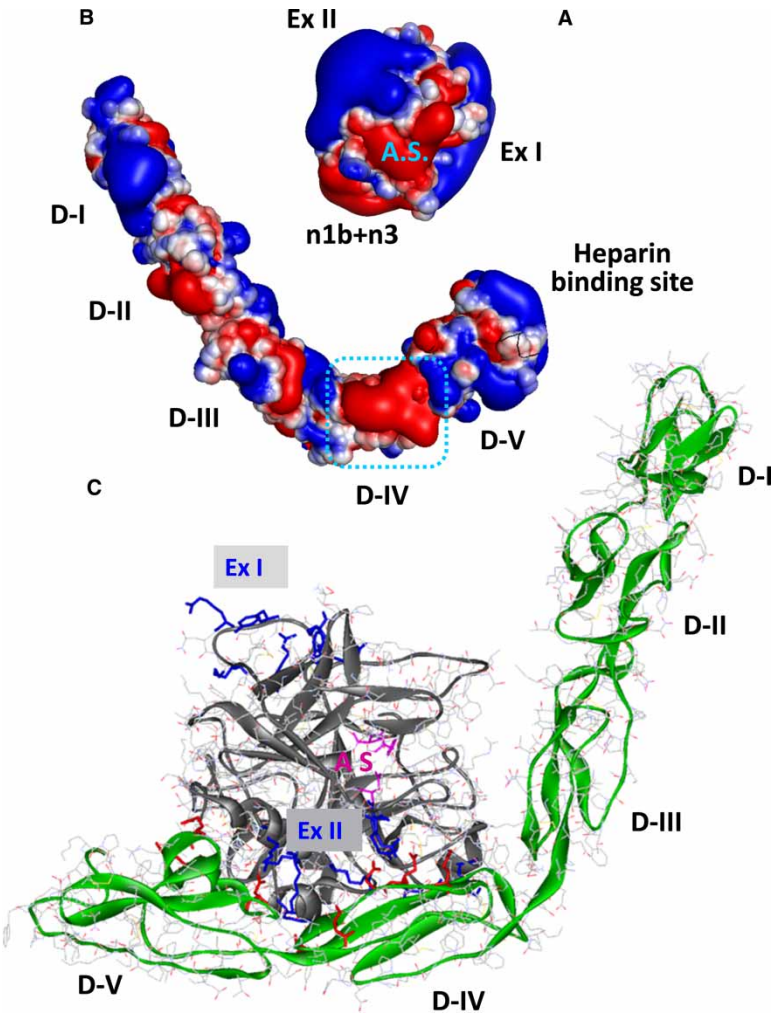


Figure 11. Theoretical model of αT–β2GpI interaction.

Surface electrostatic properties of αT (**A**) and β2GpI (**B**). The protein surfaces are color-coded according to their electrostatic potential, from −1 kcal/mol/e$^-$ (deep red) to +1 kcal/mol/e$^-$ (deep blue). Calculations were performed using the program APBS [29], run on the X-ray structure of β2GpI (1c1z) and des-PPACK αT (1ppb). (**A**) In αT structure (top view), the negatively charged active site, A.S. (H57,D102,S195), is surrounded by a 'ring' of negative amino acids (E39,D60e,E61,D63,E97a,E146,E192,E217, D221a), protruding from the protease insertion loops and flanked by the positive regions defining exosite-I (F34,K36,R67,R73, Y76,R77a,K81) and exosite-II (R93,R101,R126,R165,K169,R233,K235,K236,K240), which are positioned almost opposite to the active site. Calculations reveal the presence of an additional negative region in αT, formed by n1b + n3 patches (E14e,E14h, D14l, E164,D186a,E186b,D222) [16]. (**B**) For β2GpI (side view), a large electronegative surface can be identified in the region spanning D-IV (D217,D222,E225,E226,E228) and the upper face of D-V (E259,E265,D275), whereas the lower face of D-V, i.e. the heparin-binding site, is strongly electropositive. (**C**) Ribbon drawing of the complex formed by αT (gray) and β2GpI (green), as obtained by ClusPro 2.0 (model_01). The position of the αT active site and exosites I and II are indicated, along with some (partially) uncompensated basic amino acids (blue) in αT exosite-II (R93,R101,R126,K236) and acidic uncompensated residues (red) in β2GpI (E217,E225,E226,E275). The model suggests that the αT active site and exosite-I are not involved in β2GpI interaction and are accessible for ligand binding.

(Figure 10B). These results suggest that the two peptides and β2GpI likely cover overlapping regions on the αT surface, even though the details of ionic interactions they establish with αT might be different. This interpretation is in keeping with the data in Figure 3, showing that αT–β2GpI binding is influenced by ionic interactions, which are intrinsically unspecific in nature [35]. Indeed, charge–charge interactions are formed by long,

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flexible amino acids (Asp, Glu, Lys and Arg) that often form dynamic ionic networks on proteins, whereby the loss of binding energy caused by disruption of a specific salt bridge at the complex interface can be compensated by the formation of novel electrostatic interactions nearby, with minimal structural/energetic perturbation [35].

Following the site-specific perturbation strategy we have exploited above for mapping the binding sites on αT, we decided to probe the role of β2GpI D-V by measuring the affinity of the nicked species β2GpI* for αT, where β2GpI* is generated by proteolysis with plasmin during fibrinolysis *in vivo* and corresponds to the natural protein having the single peptide bond Lys317–Thr318 in D-V cleaved [1,4]. Notably, the electropositive face of D-V (Figure 11A) binds to negatively charged molecular targets, i.e. anionic phospholipids, heparin and cellular receptors [1,4,50,51], mainly through the positive segment ^{282}KNKEKK^{287}, connecting strands βC and βD, and the C-terminal flexible loop, ^{308}KEHSSLAFWK↓TDASDVKPC^{326}, containing the cleavage site for plasmin (↓). Notably, after proteolysis, the loop 308–326 remains linked to the protein core by the two disulfide bonds Cys281–Cys306 and Cys288–Cys326. NMR analysis of intact and nicked recombinant D-V reveals that even though peptide bond cleavage does not induce global unfolding of D-V, it locally alters the conformation of the loop 308–326, which becomes much more flexible, and strongly perturbs the electrostatic potential of D-V, due to the generation of an additional positive and negative charge at the cleaved peptide bond [52] (see Supplementary Figure S4D). As a result, β2GpI* (10–25 μg/ml) does not bind to anionic phospholipids and heparin [50]. Hence, β2GpI* can be taken as a suitable probe for studying the role of the D-V electropositive region in β2GpI-interacting systems. In the present study, β2GpI* was prepared by proteolysis of plasma β2GpI with plasmin, purified to homogeneity (>95%) according to Goto and coworkers [52] and thoroughly characterized by RP-HPLC, SDS–PAGE and HR-MS (Supplementary Figure S4A). Interestingly, the fluorescence quantum yield of β2GpI* was reduced by ∼30% compared with the intact protein (Supplementary Figure S4B), fully consistent with the interpretation that (partial) unfolding of the nicked loop 308–326 alters the conformation/environment of the single Trp316 next to the cleavage site. Importantly, fluorescence binding data in Figure 10B indicate that intact β2GpI and nicked β2GpI* bind to αT with identical affinities, thus suggesting that the phospholipid/heparin-binding site in D-V is not likely to be involved in the interaction with αT.

Modeling β2GpI–αT interaction

To interpret on structural grounds the effects of β2GpI on αT functions, we first investigated the electrostatic properties of the two proteins (Figure 11A,B) and then produced a theoretical model of the αT–β2GpI complex (Figure 11C). Electrostatic potential calculations highlight the unique and asymmetric surface electrostatics of αT [16,53], with two positively charged patches, i.e. exosites-I and -II, flanking the enzyme active site, which conversely has a strong negative potential and is surrounded by a 'negative ring' of acidic residues, also denoted as the n1a negative patch [16], in the insertion loops shaping the substrate-binding sites. Another large negative region comprises the outer surface of the loops 180 and 220 (i.e. the n1b patch), shaping the Na^+-binding site, and the C-terminal helix of the A-chain (i.e. the n3 patch) [16]. The negative electrostatic potential of this region is probably overestimated, as the software we used for calculations did not consider the contribution of the Na^+ cation bound to αT. Asymmetric distribution of charges is also common to β2GpI, where a large electronegative surface can be identified in the hook-shaped region spanning the upper face of D-IV and -V, whereas the lower face of D-V is highly electropositive [6,7]. Hence, both αT and β2GpI contain either positive or negative regions where amino acids with uncompensated charges display high electrostatic potential and thus may function as potent attractors for complex formation.

The theoretical model of the αT–β2GpI complex (Figure 11C) was generated starting from the structure of full-length β2GpI and des-PPACK αT, using the 'van der Waals + electrostatics' option, as implemented in the online version of the automated docking algorithm ClusPro 2.0 [31], without imposing any geometric/energetic constraint. Briefly, the server performs three computational steps as follows: (1) rigid-body docking; (2) hierarchical clustering to select and rank near-native docked conformations and (3) refinement of selected structures. The ranking method is based on the hypothesis that clusters of near-native structures are located in broad energy funnels, such that the most likely models of the complex are represented in the most populated clusters. As from the default output of ClusPro, the 10 top-ranked models were analyzed and the representative structures (models 00 and 01) of the two most populated clusters (containing 157 and 131 members, respectively) were considered as possible candidates for the αT–β2GpI complex (Supplementary Figure S8). Model_00 indicates that the highly electropositive lower face of D-V, i.e. the phospholipid/heparin-binding site encompassing the loop regions 282–287 and 308–326, might interact with a negative region extending over the n1b +

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n3 patches in αT (see above), while leaving the active site and both exosites unmasked. This model, however, was discarded as it contradicts our experimental data, demonstrating that αT exosite-II interacts with β2GpI (Figures 6 and 8) and, conversely, that the phospholipid/heparin-binding site in β2GpI D-V is not involved in binding to αT (Figure 10B). Hence, model_01 (representing the second most populated cluster) was selected as the best representative of the αT–β2GpI complex (Figure 11C). This model captures the shape and electrostatic complementarity existing between αT and β2GpI structures, whereby the highly electropositive exosite-II on the convex αT surface interacts with the electronegative concave surface in β2GpI, shaped by D-IV and part of D-V, while leaving the active-site region and exosite-I fully accessible for binding (see Figure 11C). This image nicely agrees with the experimental data reported in this work, highlighting the salt dependence of αT–β2GpI interaction (Figure 3) and indicating that the positive exosite-II is the hot spot on αT for binding to β2GpI (Figures 6 and 8), whereas the protease active site (Figure 5) and exosite-I (Figures 7–9) are not involved in complex formation. Consistent with the results of deglycosylation experiments (Figure 4), the model also

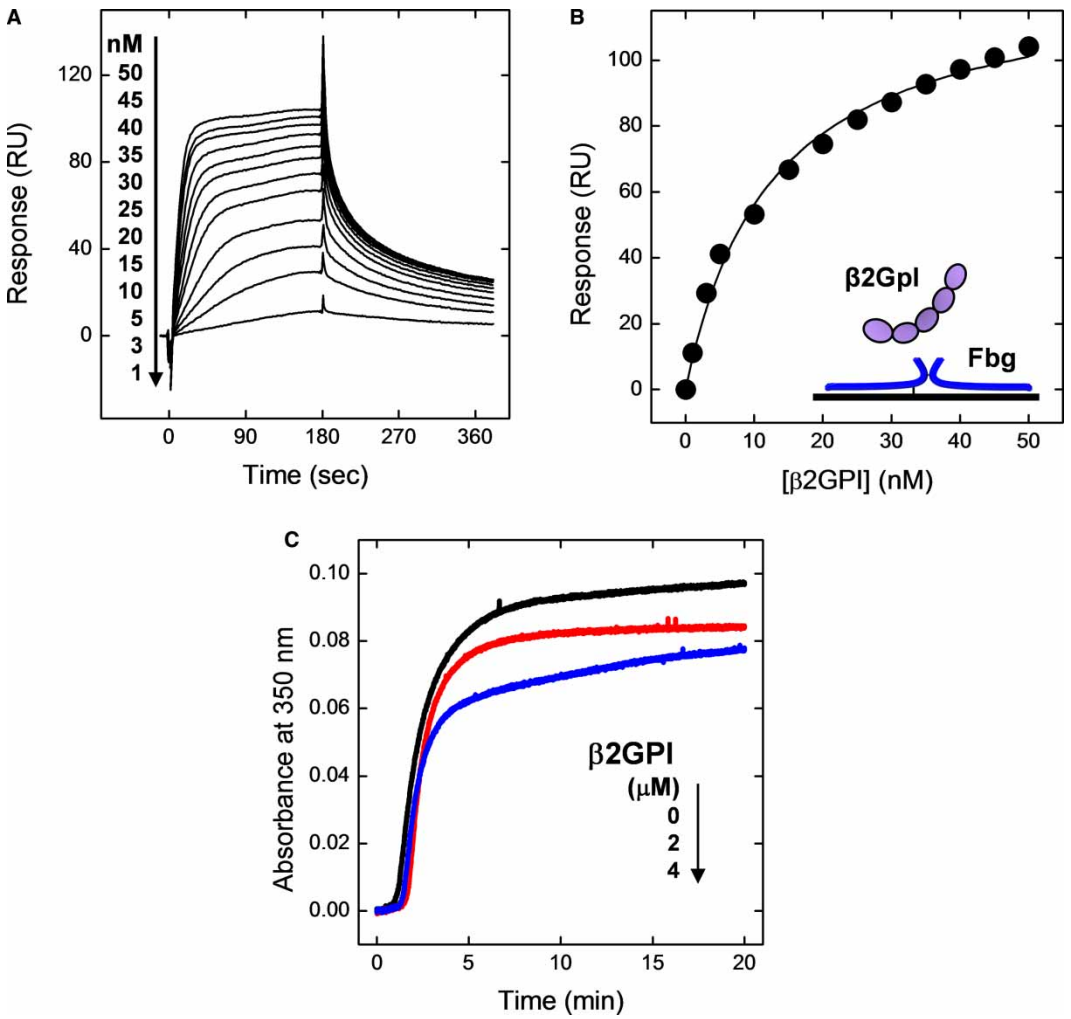


Figure 12. SPR binding of β2GpI to fibrinogen and effect of β2GpI on fibrin generation.

SPR analysis (**A** and **B**) of β2GpI–fibrinogen interaction and turbidimetric analysis of the effect of β2GpI on fibrin generation. (**A** and **B**) Binding of β2GpI to fibrinogen immobilized on a CM5 sensor chip. Sensograms (**A**) and plot (**B**) of RU_{max} versus fibrinogen concentration (●). Fitting of data points with eqn 2 in Supplementary Data yielded $K_d = 13 \pm 2$ nM. All measurements were carried out at 25°C in HBS-EP$^+$ (pH 7.4). (**C**) Effect of β2GpI of fibrin generation. Representative clotting curves for the generation of fibrin in the presence of increasing β2GpI concentrations: 0, —; 2, ▬; 4, ▬ μM. To a human fibrinogen solution (0.35 μM) in HBS (pH 7.4) at 25°C was added αT (1 nM), with or without β2GpI (4 μM), and the time course increase in turbidity (absorbance at 350 nm) was recorded.

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argues against a direct involvement of the carbohydrate chains in αT–β2GpI interaction, as these point in different directions in the complex or are too far for productive binding. Intriguingly, model_01 also seems to account for unexpected results, such as the increased affinity (∼4-fold) of the D178A αT mutant for β2GpI (Supplementary Figure S5). D178 is intramolecularly salt bridged to R165 and R233 at the periphery of exosite-II in αT structure [16] and, in the docking complex with β2GpI, it points toward the negative surface of D-IV, at a distance of 7–8 Å from the uncompensated E225 and E226 (Figure 11C). Hence, if it is assumed that αT structure is not altered by D178A mutation, it is conceivable to propose that ablation of the negative charge at position 178 might increase the affinity of αT for β2GpI by either strengthening the resulting positive electrostatic potential of the protease at the mutated site or, concomitantly, by alleviating the electrostatic repulsion of D178 with the negative region of D-IV. Of note, in models 03, 04, 08 and 09, αT assumes different orientations on β2GpI structure, but nevertheless the interacting regions we have observed in model_01 are roughly conserved.

Overall, although our model snugly fits with the experimental data here reported, it is important to emphasize that the model awaits structural verification by direct experiments, e.g. X-ray, NMR or small-angle X-ray scattering techniques.

Effect of β2GpI on fibrin structure

The lack of any relevant effect of β2GpI on fibrinogen hydrolysis (see above) suggests that the prolongation of the clotting time induced by β2GpI in our previous fibrin clotting assays [19] cannot be ascribed to the alteration in the efficiency of fibrinopeptide release. These considerations prompted us to investigate the alternative possibility that β2GpI directly binds to fibrinogen and alters the structure of the ensuing fibrin clot through a mechanism not dependent on the effect that β2GpI might exert on αT-induced hydrolysis of fibrinogen. To this aim, SPR measurements were performed by challenging a fibrinogen-bound CM5 sensor chip with incremental concentrations of β2GpI. Strikingly, our data (Figure 12A,B) indicate that fibrinogen binds to immobilized β2GpI with high affinity ($K_d = 13 \pm 2$ nM). Thereafter, the effect of β_2GpI on the time course generation of fibrin was studied by turbidimetry at 350 nm (Figure 12C), yielding different clotting curves in the absence and presence of β2GpI. Specifically, the clotting time (t_c) remained essentially constant, whereas the A_{max} value, i.e. the maximum turbidity measured when the final fibrin clot is formed, was reduced by ∼20% with 4 μM β2GpI.

Typically, a fibrin clotting curve displays: (1) a lag phase, (2) a linear rise and (3) a plateau [32]. The lag phase (1) corresponds to the time necessary for the longitudinal elongation of protofibrils deriving from fibrin monomers, after removal of fibrinopeptides; thereafter, the rapid linear rise in turbidity (2) results from lateral aggregation of those protofibrils that have reached a certain threshold length to aggregate; finally, the turbidity reaches a plateau (3) when most of the protofibrils have been transformed into fibers [32]. It is important to note that the value of A_{max} provides key geometric parameters of fibrin, as it is proportional to the square of the average diameter of the fibers [32]. In the light of these considerations, the relative invariance of the lag phase is consistent with the observation that β2GpI does not alter the rate of fibrinopeptide release (Figure 9A), whereas the decrease in A_{max} indicates that β2GpI affects the lateral aggregation of fibrin, inducing the formation of thinner fibers, which generate a less intense turbidimetric signal (Figure 12C).

Discussion

Although the involvement of β2GpI in the pathogenesis of APS is widely accepted, the physiological function of this protein is still unclear, and in the last three decades, both procoagulant and anticoagulant functions *in vitro* have been reported [4]. To accomplish these functions, β2GpI interacts with different target systems, including negatively charged membranes of activated platelets, coagulation factor XI zymogen, von Willebrand factor, plasminogen and tissue plasminogen activator, platelet factor 4 and lipoprotein receptors [54–59]. Recently, we added another piece to the puzzle of β2GpI functions [19] and showed that this protein prolongs the fibrin clotting time in ecarin clotting time (ECT) and thrombin time (TT) assays on β2GpI-deficient plasma and inhibits αT-induced platelet aggregation in several different experimental settings, including impedance aggregometry on whole blood, transmittance aggregometry on gel-filtered platelets and immunocytofluorimetry for measuring PAR1 cleavage on platelet membranes. In the same study, we demonstrated that β2GpI does not alter the efficiency with which αT activates the anticoagulant PC, regardless of TM addition [19]. Hence, we proposed that β2GpI may function as a physiological anticoagulant by interacting with αT and selectively inhibiting the procoagulant functions (i.e. normal fibrin generation and platelet aggregation) of the

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protease without affecting its unique anticoagulant activity (i.e. aPC generation). This hypothesis is in keeping with several lines of clinical evidence, whereby reduced β2GpI levels are found in patients with stroke and thrombotic disorders [60,61], whereas high circulating levels of β2GpI reduce the risk of myocardial infarction [62]. Furthermore, there is a positive correlation between the presence of autoantibodies against β2GpI and thrombotic manifestations in APS patients [1–4].

To shed light on the mechanism underlying the effects of β2GpI on αT functions, in the present study we first carefully established, by DLS and fluorescence spectroscopy (Figure 1) and SPR (Figure 2), that β2GpI forms a stable complex with αT (K_d = 50–100 nM) and then mapped the binding sites on αT (Figures 5–9) mainly using the site-specific perturbation strategy, whereby a decrease in the αT–β2GpI affinity caused by perturbation (i.e. amino acid exchange, ligand binding and proteolysis) of a given site on αT was taken as an indication that the perturbed site was directly involved in binding. The possible involvement of the αT active site in β2GpI interaction was ruled out by enzymatic activity measurements (Figure 5A), showing that physiological β2GpI concentrations did not alter the efficiency of S2238 hydrolysis by αT, and by SPR measurements (Figure 5B), indicating that neither blockage of the αT active site with PPACK nor disruption of the catalytic pocket, as in the Pre2 zymogen, compromised binding to β2GpI. The involvement of exosite-II was highlighted by SPR measurements, using the save exosite-II binder GpIbα(268–282), which was able to compete with β2GpI for binding to αT (Figure 6). Accordingly, masking of exosite-II, as in the ProT zymogen, strongly reduced the affinity of αT for β2GpI (Figure 8). Intriguingly, even exosite-I binders, such as hirugen and HD1 aptamer, competed with β2GpI in binding to αT (Figure 6B). However, this result was interpreted as arising from an indirect effect generated by the negative allosteric coupling existing between the two exosites [46–48]. This interpretation was corroborated by three lines of evidence: first, β2GpI failed to displace fluoresceinated hirugen from αT exosite-I (Figure 8); secondly, selective disruption of exosite-I by proteolysis, as in β_T-thrombin, did not alter the affinity for β2GpI (Figure 9); thirdly, β2GpI did not affect the efficiency with which αT cleaves two exosite-I specific ligands/substrates like fibrinogen and PAR1(38–60) (Figure 9). Although our data concurrently indicate that exosite-II is the hot spot on αT for β2GpI binding, some promiscuous binding at exosite-I cannot be ruled out, especially at high local concentrations of protein *in vivo* (i.e. on the activated platelets surface) or when exosite-II is occupied. Thereafter, starting from the knowledge that charge–charge interactions significantly contribute to αT–β2GpI association and that the strongly positive exosite-II of thrombin is involved in binding, we identified the negatively charged segment ^{222}**D**GP**EE**I**E**^{228} in β2GpI D-IV as a putative binding site for αT. Indeed, the corresponding synthetic peptide β2GpI(219–232) binds to αT with an affinity only 2-fold lower than that of the full-length β2GpI. Importantly, the theoretical docking model of the αT–β2GpI complex reported in Figure 11C agrees well with all the experimental data reported in the present study and sets the basis for future mutagenesis work and direct structural determination.

The proposed model also bears important implications for understanding the effect of β2GpI on αT functions. First of all, the involvement of exosite-II explains in a straightforward way how β2GpI impairs αT-mediated platelet aggregation [19]. GpIbα is responsible for anchoring αT on the platelet surface through exosite-II binding, thus orienting the protease exosite-I and active site for efficient binding to and cleavage of PAR1 [18]. Notably, αT–GpIbα interaction is the rate-limiting step in the pathway leading to platelet activation, in the sense that impairment of αT–GpIbα interaction by an interfering molecule can inhibit platelet aggregation even though the molecule does not directly influence PAR1 hydrolysis. Hence, the co-localization of αT and β2GpI on the platelet surface [1,4,11,13,17] and the competition of β2GpI and GpIbα(268–282) for binding to αT exosite-II (Figure 6) suggest that β2GpI may function as a scavenger of the protease, thus preventing αT binding to GpIbα and the resulting platelet aggregation mediated by PAR1 cleavage. The concomitant shielding of exosite-II and exposure of the αT active site and exosite-I in the β2GpI–αT complex nicely explains apparently puzzling data whereby β2GpI inhibits αT-mediated cleavage of PAR1 on intact platelets [19] without impairing PAR1(38–60) hydrolysis by αT (Figure 9B). Likewise, the accessibility of the αT active site and exosite-I in the β2GpI–αT complex is fully consistent with the observation that β2GpI does not interfere with αT-mediated hydrolysis of PC to generate aPC either in the absence or presence of TM, a cofactor that tightly binds to αT exosite-I and brings PC zymogen into close proximity for accelerating aPC generation [10].

The interpretation of the effect that β2GpI has on fibrin generation in plasma clotting assays deserves particular attention. In our previous work [19], we showed that β2GpI prolongs the clotting time, t_c, in clinical tests, such as ECT and TT, on β2GpI-deficient plasma, suggesting that the protein interferes with the

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αT-catalyzed conversion of fibrinogen into fibrin. Notably, t_c corresponds to the time necessary to increase, above a certain threshold, the turbidity signal (absorbance at 671 nm) generated by the newly formed fibrin. Nonetheless, kinetic measurements reported in Figure 9A clearly demonstrate that β2GpI does not alter the rate of fibrinopeptide release from purified fibrinogen and this is fully consistent with our structural model (Figure 11C), showing that β2GpI does not mask the reactivity of the αT active site and exosite-I, which is actually the fibrinogen-binding site on αT [11,13,17]. These findings indicate that the apparent increase in t_c, caused by β2GpI in plasma clotting assays, should not be ascribed to the inhibition of fibrinogen hydrolysis by αT and prompted us to explore the possibility that β2GpI may directly bind to fibrinogen and alter fibrin generation through a mechanism independent of αT proteolysis. Strikingly, SPR data in Figure 12A,B indicate that β2GpI tightly interacts with fibrinogen, while turbidimetric analysis provides clear-cut evidence that β2GpI dose-dependently induces the generation of fibrin fibers having a smaller diameter compared with those obtained without β2GpI (Figure 12C). Hence, the prolongation of the t_c value induced by β2GpI [19] more likely reflects the inhibition of the lateral aggregation of fibrin oligomers rather than the impairment of fibrinogen hydrolysis by αT.

In conclusion, by combining molecular modeling with biochemical/biophysical techniques, in this work, we produced a coherent structural model of the αT–β2GpI complex that allows one to rationalize at the molecular level the effects of β2GpI on αT functions. These results will contribute to elucidate the role of the multifunctional protein β2GpI in the physiology of normal coagulation and, hopefully, to understand the pathogenetic mechanisms underlying the thrombotic effects of anti-β2GpI autoantibodies in APS.

Abbreviations

%Pd, percent polydispersity; αT, α-thrombin; αT-PPACK, PPACK-inhibited αT; β_TT, β_T-thrombin; [F]-hirugen, *N*α-fluoresceinated hirugen; aPC, active protein C; APS, antiphospholipid syndrome; CaM, calmodulin; CC, counter-ion condensation; d-β2GpI, deglycosylated β2GpI; D_H, hydrodynamic diameter; DLS, dynamic light scattering; ECT, ecarin clotting time; FpA, fibrinopeptide A; FpB, fibrinopeptide B; GpIbα, platelet receptor glycoprotein Ibα; HBS, 20 mM HEPES (pH 7.4) and 0.15 M NaCl; HBS-EP$^+$, HBS containing 3 mM EDTA and 0.05% polyoxyethylene sorbitan; HEPES, 2-[4-(2-hydroxyethyl)piperazin-1-yl]ethanesulfonic acid; HR-MS, high-resolution mass spectrometry; MS, mass spectrometry; PAR1, type 1 protease-activated receptor; PC, protein C; PNGase-F, peptide-*N*-glycosidase F; PPACK, (D)-phenylalanyl-prolyl-arginyl chloromethyl ketone; Pre2, recombinant prethrombin 2 zymogen expressed in *E. coli*; ProT, prothrombin; R-β2GpI(219–232), randomized peptide sequence 219–232 of β2GpI; RU, response units; SPR, surface plasmon resonance; TLCK, tosyl-lysyl chloromethyl ketone; TM, thrombomodulin; TT, thrombin time; β2GpI, β2 Glycoprotein I.

Author Contribution

L.A., D.P., S.T. and N.P. performed research; V.P. provided human plasma samples; V.D.F. designed the research, analyzed and interpreted the results, and wrote the manuscript; all authors reviewed the content of the manuscript.

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Competing Interests

The Authors declare that there are no competing interests associated with the manuscript.

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